

Project Title:

Healthcare Medical Chatbot Using Artificial Intelligence and NLP

Presented by : Batch – 24

Location: VFSTR-Vadlamudi, Andhra Pradesh, India

The Internal Departmental Project (IDP) review was conducted to evaluate our practical knowledge, technical skills, and project progress, where we, the students of Batch-24, presented our project titled “Healthcare Medical Chatbot” in a professional environment before faculty evaluators. The main objective of our project is to develop an intelligent AI-powered healthcare platform that assists patients in understanding their symptoms, obtaining preliminary diagnosis results, and connecting with appropriate healthcare professionals through an interactive web-based chatbot system. By utilizing Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) techniques, the system analyzes patient symptoms entered in natural language and predicts possible diseases with confidence scores and specialist recommendations.

Our project integrates modern frontend and backend technologies, enabling users to interact with the chatbot through a responsive and user-friendly interface while the backend processes medical data using a fine-tuned BERT-based NLP model. The system includes important features such as patient registration and login, chatbot-based symptom analysis, disease prediction results, doctor recommendations, appointment booking, appointment history, and separate dashboards for doctors and administrators. During the review, we explained the project objectives, software requirements, system architecture, database schema, NLP pipeline, UML diagrams, and implementation methodology. We also demonstrated the working model of the application, including chatbot interaction, disease prediction functionality, appointment scheduling, and management modules.

Each team member actively presented different modules such as frontend development using React.js, backend API development using Flask, database management using MySQL, AI/NLP model integration using Hugging Face Transformers and BERT, and cloud deployment using AWS infrastructure. The evaluators reviewed our documentation, UI design, system workflow, testing methodologies, and implementation process, and appreciated the successful integration of AI technologies with a full-stack healthcare application. Valuable suggestions were provided regarding improving prediction accuracy, enhancing data security, reducing response time, increasing dataset quality, and implementing multilingual support for wider accessibility.

During the development phase, we faced several challenges including collecting and preprocessing medical datasets, training and integrating the BERT NLP model, optimizing chatbot response performance, managing secure authentication, and deploying the system on cloud infrastructure. These challenges were successfully overcome through continuous research, experimentation, teamwork, testing, and problem-solving approaches. In the future, the project can be enhanced by integrating telemedicine video consultation, wearable device support,

multilingual communication, mobile application development, real-time health monitoring, and expanded disease datasets for improved diagnosis accuracy.

